

What Actually Works in Agentic Knowledge Graph Question Answering: A Component-Level Attribution Study

Md. Sakif Khan^a, Sadia Sharmin^a

^aDepartment of Computer Science and Engineering, Bangladesh University of Engineering and Technology, Dhaka, Bangladesh

Abstract

Keywords: knowledge graph question answering, large language models, agentic retrieval, ablation study, hallucination, benchmark quality

Email address: sakifkhanopu@gmail.com (Md. Sakif Khan)

1. Introduction

2. Related Work

3. The AGR Framework

3.1. State Machine and Tool API

3.2. Structural Verification Layer

3.3. Budgets and Termination

4. Experimental Setup

4.1. Knowledge Environment

4.2. Benchmarks, Baselines, and Backbone

4.3. Metrics and Protocol

5. Results

5.1. Accuracy

5.2. Cost and the Location of the Margin

5.3. Accuracy by Hop Depth

5.4. Groundedness

6. Component-Level Attribution

6.1. When Planning Hurts

6.2. Three Components with No Detected Effect

7. Error Analysis

7.1. The Echo Attractor

7.2. Benchmark Label Defects

8. Discussion and Limitations

9. Conclusion

Declaration of competing interest

3

The authors declare no competing financial interests or personal relationships that could have influenced the work reported here.

Data availability

Code, prompts, per-question run records, and the analysis scripts that produce every figure and table in this paper are available at <https://github.com/sakif-khan/agentik-graph-reasoning> under the MIT licence. The knowledge environment is derived from Freebase [1] and is rebuilt from its public source by the scripts in that repository rather than redistributed; the question sets are the public WebQSP [2] and ComplexWebQuestions [3] releases.

References

- [1] K. Bollacker, C. Evans, P. Paritosh, T. Sturge, J. Taylor, Freebase: A collaboratively created graph database for structuring human knowledge, in: Proceedings of the 2008 ACM SIGMOD International Conference on Management of Data, 2008, pp. 1247–1250.
- [2] W.-t. Yih, M. Richardson, C. Meek, M.-W. Chang, J. Suh, The value of semantic parse labeling for knowledge base question answering, in: Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers), 2016, pp. 201–206.
- [3] A. Talmor, J. Berant, The web as a knowledge-base for answering complex questions, in: Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, 2018, pp. 641–651.